Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT JEE-MAIN Date: 31-12-2025

Time: 09.00Am to 12.00Pm GTM-07 Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**2**	2	**4**	3	**4**	4	**2**	5	**3**
6	**3**	7	**3**	8	**3**	9	**1**	10	**1**
11	**2**	12	**3**	13	**4**	14	**3**	15	**2**
16	**4**	17	**2**	18	**3**	19	**4**	20	**4**
21	**6**	22	**2**	23	**6**	24	**6**	25	**1**

PHYSICS

26	**3**	27	**3**	28	**2**	29	**1**	30	**2**
31	**3**	32	**1**	33	**1**	34	**1**	35	**4**
36	**2**	37	**4**	38	**2**	39	**1**	40	**3**
41	**2**	42	**3**	43	**1**	44	**4**	45	**1**
46	**9**	47	**4**	48	**4**	49	**15**	50	**4**

CHEMISTRY

51	**3**	52	**3**	53	**4**	54	**3**	55	**4**
56	**1**	57	**4**	58	**4**	59	**4**	60	**3**
61	**4**	62	**3**	63	**2**	64	**1**	65	**4**
66	**4**	67	**4**	68	**2**	69	**2**	70	**1**
71	**24**	72	**300**	73	**4**	74	**15**	75	**318**

SOLUTION

MATHEMATICS

1. The point B is $(2,1)$

(B is image of A about line $y = x$)

Image of $A(1,2)$ in the line $x-2y+1=0$ is

Given by $\frac{x-1}{y}=\frac{y-2}{-2}=\frac{4}{5}$ $\therefore$ coordinates of the point are $\left(\frac{9}{5},\frac{2}{5}\right)$

Since this point lies on BC.

$\therefore$ equation of BC is $3x-y-5=0$ $\quad \therefore a+b=2$

2. An interior point of the smaller region is interior point of the parabola as well as of the circle. $\therefore (m+1)^2-4(-2m)<0$

And $4m^2+(m+1)^2-4<0$

i.e $m^2+10m+1<0$

and $5m^2+2m-3<0$

ie. $\frac{-10-\sqrt{96}}{2}<m<\frac{-10+\sqrt{100-4}}{2}$ and $\frac{-2-8}{10}<m<\frac{-2+\sqrt{4+60}}{10}$

ie. $-5-2\sqrt{6}<m<-5+2\sqrt{6}$

and $-1<m<\frac{3}{5}$ ie. $-1<m<2\sqrt{6}-5$

3. $(P)\, f(x)=|x^3|$ is continuous and differentiable

$(Q)\, f(x)=\sqrt{|x|}$ is continuous

$f'(x)=\frac{1}{2\sqrt{|x|}}.\frac{x}{|x|}$ {does not exist at $x=0$}

$(R)\, f(x)=|\sin^{-1}x|$ is continuous $\quad f'(x)=\frac{\sin^{-1}x}{|\sin^{-1}x|}.\frac{1}{\sqrt{1-x^2}}$

{does not exist at x=0}

$(S)\, f(x)=\cos^{-1}|x|$ is continuous

$f'(x)=\frac{-1}{\sqrt{1-x^2}}.\frac{x}{|x|}$ {does not exist at x=0}

4. Given condition $A^2-B^2=(A-B)(A+B)$

$\Rightarrow A^2-B^2=A^2+AB-BA-B^2$ $\quad \Rightarrow AB=BA$.

5. Term independent of x will be ${}^{10}C_5(\sin\alpha)^5(\cos\alpha)^5=\frac{{}^{10}C_5}{2^5}\sin^5(2\alpha)$

So maximum value will be $=\frac{{}^{10}C_5}{2^5}$

6. In the neighbourhood of x = 0, f(x) = log 2 – sin x

$\therefore$ g(x) = f(f(x)) = log2 – sin (f(x))

log 2 – sin(log 2 – sin x)

Since g(x) is differentiable at x = 0,

$\therefore g'(x) = -\cos(\log 2 - \sin x)(-\cos x)$ $\quad \Rightarrow g'(0)=\cos(\log 2)$

7. $f(x)+3g(x)=x^2+x+6$ .....(i)

$f(x)+2g(x)=x^2+2$ .....(ii)

Hence $g(x)=x+4$

$\therefore f(x)=x^2-2x-6$

Now $g(\tan^2 x)-f((\tan x)-8)+(4+\tan^2 x)-(\tan^2 x-2\tan x-6)-8=2\tan x+2$

$$J=\int_0^{\pi/4}\ln(g\tan^2 x)-f(\tan x)-8dx=\int_0^{\pi/4}\ln(2\tan x+2)dx$$

$$J=\int_0^{\pi/4}\ln(2\tan x+2)dx.....(1)$$

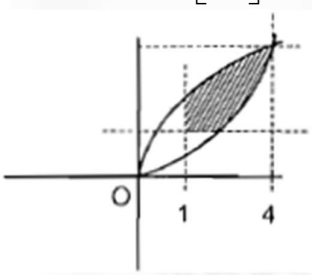
Using property

$$J=\int_0^{\pi/4}\ln\left(\frac{4}{1+\tan x}\right)dx.....(2)$$

Adding equation (1) &(2) we get $J=\frac{3\pi}{8}\ln 2$

8. For $y\in(1,4):\left[\sqrt{y}\right]=1, C_1: y^2=4x$

For $X\in(1,4);\left[\sqrt{x}\right]=1, C_2: x^2=4y$



Area $=\int_1^2\left(2\sqrt{x}-1\right)dx+\int_2^4\left(2\sqrt{x}-\frac{x^2}{4}\right)dx=\frac{11}{3}sq.units$

9. $y=ax^2$ $\Rightarrow\frac{dy}{dx}=2ax\Rightarrow a=\frac{1}{2x}\frac{dy}{dx}$

Now equation of curve $y=\frac{1}{2x}\frac{dy}{dx}x^2$ $\Rightarrow\frac{x}{2}\frac{dy}{dx}=y$

Orthogonal trajectory $-\frac{dx}{dy}=\frac{2y}{x}$ $\frac{x^2}{2}=-y^2+c\Rightarrow y^2+\frac{x^2}{2}=c$

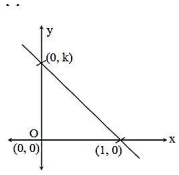
This is the equation of ellipse

10. $\alpha+\beta=p,\alpha\beta=-(p+q)\Rightarrow(\alpha+1)(\beta+1)=1-q$

$$\frac{(\alpha+1)^2}{(\alpha+1)^2+q-1}+\frac{(\beta+1)^2}{(\beta+1)^2+q-1}=\frac{2(1-q)^2+(q-1)\left((\alpha+1)^2+(\beta+1)^2\right)}{2(1-q)^2+(q-1)\left((\alpha+1)^2+(\beta+1)^2\right)}=1$$

11.

Sol.



Area $=\frac{1}{2}h.k=50$

$h.k=100$

$h.k=2^2.5^2$

Total divisors $=(2+1)(2+1)=a$

If $h>0, k>0$

But $\left.\begin{matrix} h>0, & k<0 \\ h<0, & k>0 \\ h<0, & k<0 \end{matrix}\right\}$ all are possible so that total no. of positive case $9+9+9+9=36$

12. $\Delta=\begin{vmatrix} 1 & 1 & 1 \\ 1 & a & 1 \\ a & b & 1 \end{vmatrix}=0 \qquad -(a-1)^2=0 \Rightarrow a=1$

For b=1 only the system has no solution.

13. $a_1^r a_2^k = a^r(aR)^k = a^{r+k}R^k \qquad a_2^r a_3^k = (aR)^r(aR^2)^k = a^{r+k}R^{r+2k} \qquad a_3^r a_1^k = (aR^2)^r(aR^3)^k = a^{r+k}R^{2r+3k}$

$$\Rightarrow \begin{vmatrix} \ln(a^{r+k}R^k) & \ln(a^{r+k}R^{r+2k}) & \ln(a^{r+k}R^{2r+3k}) \\ \ln(a^{r+k}R^{3r+4k}) & \ln(a^{r+k}R^{4r+5k}) & \ln(a^{r+k}R^{5r+6k}) \\ \ln(a^{r+k}R^{6r+7k}) & \ln(a^{r+k}R^{7r+8k}) & \ln(a^{r+k}R^{8r+9k}) \end{vmatrix}=0 \; R_2\to R_2-R_1, R_3\to R_3-R_2$$

$$\Rightarrow \begin{vmatrix} \ln(a^{r+k}R^k) & \ln(a^{r+k}R^{r+k}) & \ln(a^{r+k}R^{2r+3k}) \\ \ln\left(\frac{R^{3r+4k}}{R^k}\right) & \ln\left(\frac{R^{4r+5k}}{R^{r+2k}}\right) & \ln\left(\frac{R^{5r+6k}}{R^{2r+3k}}\right) \\ \ln\left(\frac{R^{6r+7k}}{R^{3r+4k}}\right) & \ln\left(\frac{R^{7r+8k}}{R^{4r+5k}}\right) & \ln\left(\frac{R^{8r+9k}}{R^{5r+6k}}\right) \end{vmatrix}=0 \Rightarrow \begin{vmatrix} \ln(a^{r+k}R^k) & \ln(a^{r+k}R^{r+k}) & \ln(a^{r+k}R^{2r+3k}) \\ \ln(R^{3r+3k}) & \ln(R^{3r+3k}) & \ln(R^{3r+3k}) \\ \ln(R^{3r+3k}) & \ln(R^{3r+3k}) & \ln(R^{3r+3k}) \end{vmatrix}=0$$

14. $\Delta_1=\begin{vmatrix} 1 & \sin\alpha & \cos\alpha \\ 1 & \cos\alpha & \sin\alpha \\ 1 & -\sin\alpha & \cos\alpha \end{vmatrix} = \begin{vmatrix} 0 & \sin\alpha-\cos\alpha & \cos\alpha-\sin\alpha \\ 0 & \cos\alpha+\sin\alpha & \sin\alpha-\cos\alpha \\ 1 & -\sin\alpha & \cos\alpha \end{vmatrix}$

$=(\sin\alpha-\cos\alpha)^2-(\cos^2\alpha-\sin^2\alpha)$

$=\sin^2\alpha+\cos^2\alpha-2\sin\alpha.\cos\alpha-\cos^2\alpha+\sin^2\alpha$

$=2\sin^2\alpha-2\sin\alpha.\cos\alpha$

Now, $\sin\alpha-\cos\alpha=0$ for only

$\alpha=\frac{\pi}{4} in\left(0,\frac{\pi}{2}\right) \qquad \therefore \Delta_1=2(\sin\alpha)\times 0=0,$

Since value of $\sin\alpha$ is finite for $\alpha\in\left(0,\frac{\pi}{2}\right)$

Hence non-trivivial solution for only one value of α in $\left(0,\frac{\pi}{2}\right)$

$$\begin{vmatrix} \cos\alpha & \sin\alpha & \cos\alpha \\ \sin\alpha & \cos\alpha & \sin\alpha \\ \cos\alpha & -\sin\alpha & -\cos\alpha \end{vmatrix}=0 \Rightarrow \begin{vmatrix} 0 & \sin\alpha & \cos\alpha \\ 0 & \cos\alpha & \sin\alpha \\ 2\cos\alpha & -\sin\alpha & -\cos\alpha \end{vmatrix}=0$$

$\Rightarrow 2\cos\alpha\left(\sin^2\alpha-\cos^2\alpha\right)=0 \therefore \cos\alpha=0$ or $\sin^2\alpha-\cos^2\alpha=0$

But $\cos\alpha=0$ not possible for any value of $\alpha\in\left(0,\frac{\pi}{2}\right)$

$\therefore \sin^2\alpha-\cos^2\alpha=0$

$\sin\alpha=-\cos\alpha$, *has no solution in* $\alpha\in\left(0,\frac{\pi}{2}\right)$

which is also not possible for any value of $\alpha\in\left(0,\frac{\pi}{2}\right)$

Hence, there is no solution.

15. Since a_1, a_2, a_3 are in G.P $\therefore a_2=a_1r, a_3=a_1r^2$

$\Rightarrow 9a_1+5a_1r^2>14a_1r \Rightarrow 5r^2-14r+9>0$

$\Rightarrow (5r-9)(r-1)>0 \Rightarrow r\in(-\infty,1)\cup\left(\frac{9}{5},\infty\right) \qquad \Rightarrow r\notin\left[1,\frac{9}{5}\right]$

16. $S_1=1!=1$

$1!+2!=3$

$1!+2!+3!=9$

$1!+2!+3!+4!=33$

$1!+2!+3!+4!+5!=153$

Further in every number unit digit is 3 so it can't be a perfect square.

Hence only possible values are x=1 or x=3

S_2 : Total –both together

${}^{11}C_5-{}^{9}C_3=378 \qquad S_2$: 1 2 3 4 *or* $5\times9-x$

Hence $12\times2\times10=240$

S_4 : (i) one digit number of $3'S=1$.

(ii) Two digit number

One 3's _3+3_=8+9=17

Two 3's 3 3=2

$\therefore$ number of 3's in two digit number is 17+2 =19

(iii) Three digit numbers

One 3's 3__+_3_+__3=9.9+8.9+8.9=225

Two 3's 33_+3+3+_3 3=2(9+9+8)=26.2=3.1=3

Number of 3's in three digit numbers is 225+52+3=280

Hence answer is 1+19+280=300

17. Mean $=np=4$

Variance $=npq=3 \therefore n=16, q=\frac{3}{4}, p=\frac{1}{4}$

So $P(X\geq1)=1-P(X=0)=1-\left(\frac{3}{4}\right)^{16}$

18. $\Delta(x)=\begin{vmatrix}\cos2x & \sin^2x & \cos4x\\ \sin^2x & \cos2x & \cos^2x\\ \cos4x & \cos^2x & \cos2x\end{vmatrix}$

$$\Delta(0)=\begin{vmatrix}1&0&1\\0&1&1\\1&1&1\end{vmatrix}=\begin{vmatrix}1&0&1\\0&1&1\\0&1&0\end{vmatrix}=(-1)=a_0$$

Now $\dfrac{d\Delta(x)}{d(\sin x)}=\dfrac{d(\Delta x)}{dx}/\left(\dfrac{d\sin x}{sx}\right)=\dfrac{1}{\cos x}\left\{\dfrac{d}{dx}(\Delta(x))\right\}$

At $x=0,\cos x=1$

$$\Delta(x)=\begin{vmatrix}1-2\sin^2 x & \sin^2 x & 1+8\sin^4 x-8\sin^2 x\\ \sin^2 x & 1-2\sin^2 x & 1-\sin^2 x\\ 1+8\sin^4 x-8\sin^2 x & 1-\sin^2 x & 1-2\sin^2 x\end{vmatrix}$$

Diff w.r.t to $\sin x(\sin x=t)$ at $x=0$ gives

$$\begin{vmatrix}0&0&0\\0&1&1\\1&1&1\end{vmatrix}+\begin{vmatrix}1&0&1\\0&0&0\\1&1&1\end{vmatrix}+\begin{vmatrix}1&0&1\\0&1&1\\0&0&0\end{vmatrix}=0=a_1$$

$$\frac{d}{dt}\{\Delta(x)\}_{(at\ x=0)}=\begin{vmatrix}-4t & 2t & 32t^3-16t\\ t^2 & 1-2t^2 & 1-t^2\\ 1+8t^4-8t^2 & 1-t^2 & 1-2t^2\end{vmatrix}$$

$$+\begin{vmatrix}1-2t^2 & t^2 & 1+8t^4-8t^2\\ 2t & -4t & -2t\\ 1+8t^4-8t^2 & 1-t^2 & (1-2t^2)\end{vmatrix}+\begin{vmatrix}1-2t^2 & t^2 & 1+8t^4-8t^2\\ t^2 & 1-2t^2 & 1-t^2\\ (32t^3-16t) & -2t & -4t\end{vmatrix}\qquad a_1=0$$

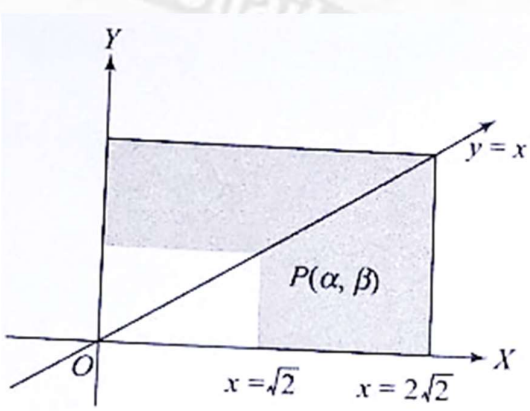
19. $(1+x)^n=1\qquad Z=e^{\left\{\frac{2n}{n}\right\}}-1$ (take $\theta=2\pi/n$)

$$Arg\left(\frac{e^{i0}-\sqrt{2}i}{\sqrt{2}}\right)=\frac{3\pi}{4}\ \frac{(\sin\theta-\sqrt{2})}{\cos\theta}=-1$$

$$\sin\theta+\cos\theta=\sqrt{2}\ \sin\left(\theta+\frac{\pi}{4}\right)=\sin\frac{\pi}{2}\ \frac{2\pi}{n}+\frac{\pi}{4}=\frac{\pi}{2}\Rightarrow\frac{2\pi}{n}=\frac{\pi}{4}\ n=8$$

20. $P_1=P_2=0, P_2=P_3=0$ and $P_3=P_1=0$ are lines of intersection of the three planes P_1, P_2 and P_3. As $\vec{n_1},\vec{n_2}$ and $\vec{n_3}$ are non coplanar, planes P_1, P_2 and P_3 will intersect at unique point. So the given lines will pass through a fixed point.

21.



Given $2\le d_1(P)+d_2(P)\le 4$

$\Rightarrow 2 \le \left|\frac{\alpha-\beta}{\sqrt{2}}\right| + \left|\frac{\alpha+\beta}{\sqrt{2}}\right| \le 4$

$\Rightarrow 2 \le |\alpha-\beta| + |\alpha+\beta| \le 4\sqrt{2}$

$\Rightarrow 2\sqrt{2} \le 2\alpha \le 4\sqrt{2}$, when $\alpha > \beta$, for $P(\alpha,\beta)$

$\Rightarrow \sqrt{2} \le \alpha \le 2\sqrt{2}$.

Area of the region $= \left(2\sqrt{2}\right)^2 - \left(\sqrt{2}\right)^2 = 8-2=6$

22. $RHL = \lim_{h\to 0} \frac{\left(\frac{\pi}{2} - \sin^{-1}\left(1-h^2\right)\right)\sin^{-1}(1-h)}{\sqrt{2}\left(h-h^2\right)} = \lim_{h\to 0} \frac{\cos^{-1}\left(1-h^2\right)\sin^{-1}(1-h)}{\sqrt{2}h\left(1-h^2\right)} = \lim_{h\to 0} \frac{\sin^{-1}\sqrt{2h^2-h^4}.\sin^{-1}(1-h)}{\sqrt{2}h(1+h)(1-h)}$

$= \lim_{h\to 0} \frac{\sin^{-1}\left(h\sqrt{2-h^2}\right)\sin^{-1}(1-h)}{\sqrt{2}h(1+h)(1-h)} = \lim_{h\to 0} \frac{1}{\sqrt{2}}.\frac{\sin^{-1}\left(h\sqrt{2-h^2}\right)}{h.\sqrt{2-h^2}}.\frac{\sqrt{2-h^2}}{1+h}.\frac{\sin^{-1}(1-h)}{1-h}$

$= \frac{1}{\sqrt{2}}.\sqrt{2}.\frac{\sin^{-1}(1)}{1} = \frac{\pi}{2} \Rightarrow K = \frac{\pi}{2}.....(1)$

$LHL = \lim_{h\to 0} \frac{A\sin^{-1}\left(1-(1-h)\right).\cos^{-1}\left(1-(1-h)\right)}{\sqrt{2(1-h)}.\left(1-(1-h)\right)}$

$\lim_{h\to 0} \frac{A.\sin^{-1}(h).\cos^{-1}(h)}{\sqrt{2}(1-h).h} = \frac{A.\pi/2}{\sqrt{2}} \Rightarrow \frac{A\pi}{2\sqrt{2}} = \frac{\pi}{2} \Rightarrow A = \sqrt{2}$

23. $g(x) = f\left(-x + f\left(f(x)\right)\right)$

$g'(x) = f'\left(-x + f\left(f(x)\right) \times \left(-1 + f'\left(f(x)\right) \times f'(x)\right)\right)$

$g'(0) = 6$

24. Given $f^3(x) = \int_0^x t.f^2(t)\,dt$

Differentiating $3f^2(x)f'(x) = xf^2(x)$

$f(x) \ne 0 \therefore f'(x) = \frac{x}{3}; \therefore f(x) = \frac{x^2}{6} + C$

But $f(0) = 0 \Rightarrow C = 0$

$f(6) = 6$

25. $\lim_{n\to\infty} \sum_{k=0}^{n} \frac{{}^nC_k}{n^k(k+3)} = \lim_{n\to\infty} \sum_{k=0}^{n} \frac{1}{k+3}\,{}^nC_k.\frac{1}{n^k}$

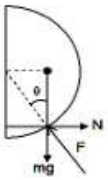
$\lim_{n\to\infty} \sum_{k=0}^{n} {}^nC_k.\frac{1}{n^k}\int_0^1 x^{k+2} \left\{\because \int_0^1 x^{k+2}dx = \frac{1}{k+3}\right\}$

$= \int_0^1 \left(x^2 \lim_{x\to\infty} \sum_{k=0}^{n} {}^nC_k.\left(\frac{x}{n}\right)^k\right)dx = \int_0^1 x^2 \left\{\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n\right\}dx = \int_0^1 x^2.e^x dx \qquad \left\{\because \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n = e^x\right\}$

$= \left(x^2.e^x\right)_0^1 - \int_0^1 2xe^x dx = e-2$

PHYSICS

26.



For equilibrium in this case all the forces should be passing through same Point. $\therefore \tan\theta = \dfrac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \dfrac{1}{\sqrt{3}}$

27. $P_0 + h\rho_1 g - \rho_2 gh + \dfrac{2T}{r} = P_0, T = \dfrac{r}{2}(\rho_2 - \rho_1)gh$

28. Given that $K = as^2$ or $\dfrac{1}{2}mv^2 = as^2$ or $mv^2 = 2as^2$

……..(i) Differentiating w.r.t time t,m $2v\dfrac{dv}{dt} = 2a.2s.\dfrac{ds}{dt}$ But, $\dfrac{ds}{dt} = v$

$\therefore 2m\dfrac{dv}{dt} = 4as$ or $m\dfrac{dv}{dt} = 2as$ Now, $m\dfrac{dv}{dt}$ =tangential force $= F_t \therefore F_t = 2as$

……(ii) Centripetal force $= F_t = \dfrac{mv^2}{R} = \dfrac{2as^2}{R}$.....(iii)

$\therefore F = \sqrt{F_t^2 + F_t^2} = \sqrt{(2as)^2 + \left(\dfrac{2as}{R}\right)^2} = 2as\sqrt{1 + \dfrac{s^2}{R^2}}$

29. (A) As both blocks moving together so

Time period $= 2\pi\sqrt{\dfrac{m}{K}}$; where m = M + m $T = 2\pi\sqrt{\dfrac{M+m}{K}}$

(B) Let block is displaced by x in (+ve) direction so force on block will be in(–ve) direction

$F = -Kx \quad (M+m)a = -Kx \quad a = -\dfrac{Kx}{(M-m)}$

(C) As upper block is moving due to friction thus $f = ma = \dfrac{mKx}{(M+m)}$

(D) This option is like two block problem in friction for maximum amplitude, force on block is also maximum, for which both blocks are moving together.

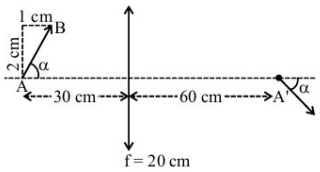
$KA = (M+m)a \quad a = \dfrac{KA}{(M+m)} \quad f = ma = \dfrac{mKA}{(M+m)} \quad f_{max} = f_L \mu mg \quad f = \mu mg \quad \dfrac{mKA}{(M+m)} = \mu mg$

$A = \dfrac{\mu(M+m)g}{K}$

(E) Maximum friction can be μmg as force is acting between blocks & normal force here

is mg.

30.



Location of image of A :- $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}\Rightarrow\frac{1}{v}-\frac{1}{-30}=\frac{1}{20}\Rightarrow\frac{1}{v}=\frac{1}{60}\Rightarrow v=60\text{cm}$

$\therefore$ m = 2 Since size of object is small wrt the location hence

$dv=m^2du\Rightarrow dv=4\times1=4\text{cm}$

$h_i=mh_0\Rightarrow h_i(dy)=2\times2=4\text{cm}$

$\therefore$ Angle made with principle axis = –45°

31. Here $5\frac{\lambda}{4}=42+0.3\times10\Rightarrow\lambda=36\text{cm}$

$\therefore$ pressure amplitude at middle of pipe is

$\Delta P=\Delta P_0\,|\sin Kx|=\Delta P_0\,|\sin\left(\frac{2\pi}{36}\times24\right)|\;=\frac{\sqrt{3}\Delta P_0}{2}$

32. Given $C=C_v+\alpha T^2\;C_v+\frac{RT}{V}\frac{dV}{dT}=C_v+\alpha T^2$

$\int\frac{\alpha T}{R}dT=\int\frac{dv}{V}+lnk\quad\frac{\alpha T^2}{2R}=ln(kV)$

$kV=e^{\frac{\alpha T^2}{2R}}\qquad\therefore Ve^{-\left(\frac{\alpha T^2}{2T}\right)}=\text{cons tan t}$

33. $F_y=mg(-\hat{j})\;F_x=\left[\frac{\sigma}{2\varepsilon_0}\right]q(\hat{i})$

$a_y=-g(\hat{i})=a_x=\left(\frac{\sigma}{m2\varepsilon_0}\right)\hat{i}$

As $S_y=H_{max},V_y=0$

Let the time taken to reach max. height be

$t_1.v_y=u_y+a_yt_1\Rightarrow0\Rightarrow u\sin\theta-gt_1\Rightarrow t_1\frac{u\sin\theta}{g}$

$At=t=t_1,v_x=u_x+a_xt_1\Rightarrow v_x=u\cos\theta+\left[\frac{\sigma q}{2m\varepsilon_0}\right]\frac{u\sin\theta}{g}$

$=u\sin\theta\left[1+\frac{\sigma q\tan\theta}{2m\varepsilon_0 g}\right]$

34. $\frac{1}{L_p}=\frac{1}{L}+\frac{1}{L}=\frac{2}{L}\Rightarrow L_p=\frac{L}{2}$

Where L is inductance of each part, $=\frac{1.8\times10^{-4}}{2}=0.9\times10^{-4}\text{H}$

$\therefore L_p=\frac{L}{2}=\frac{0.9\times10^{-4}}{2}=0.45\times10^{-4}\text{H}$

Resistance of each part, $r = \frac{6}{2} = 3\Omega$ Now, $\frac{1}{rp} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$

Time constant of circuit $, = \frac{L_p}{rp} = \frac{0.45 \times 10^{-4}}{1.5} = 3 \times 10^{-5} s$

35. Solution: Equivalent external resistance of the given circuit $R_{eq} = 4\Omega$

Current given by the cell $i = \frac{E}{R_{eq} + r} = 2A$

$(V_A - V_B) = \frac{i}{2} \times (R_2 - R_1) = -2V$

36. $E \propto \frac{Z}{n^2}$

$Z_H = 1 \quad Z_{He^+} = 2 \quad Z_{Li^{+2}} = 3$

1st excited state $\Rightarrow$ n = 2

2nd excited state $\Rightarrow$ n = 3

37. A : Since polar dielectrics are randomly oriental $\vec{P}_{net} = \vec{0}$.

R : If $\vec{E}$ is absent, polar dielectric remain polar & are randomly oriented.

38. $V_s = \frac{hv - \phi}{e}$ so stopping potential doesn't depend on Intensity $I = \frac{\eta hv}{A}$

On increasing intensity no. of photons per sec. n increases so the no. of electrons.

39. $KE = \frac{1}{2} m \left(\frac{2Gm}{R} \right) = mgR$ Assertion wrong at ∞ $U = 0$ $\therefore$ Reason correct.

40. (A) n type semiconductor holes are minority carriers and e^- are majority carriers

(B) Dopant are pentavalent atom.

(C) $n_e n_h = n_i^2$ n for intrinsic semiconductor

(E) In n type semiconductor primary source of holes generation are thermal excitation.

41. Conceptual

42. Energy of H-like atom $E = -13.6 \frac{Z^2}{n^2}$

$\therefore E \alpha Z^2 (P) \rightarrow (5)$

(Q) Energy of characteristic X-ray by Moseley's correction

$E = -13.6 (Z-1)^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$

Or, $E \alpha (Z-1)^2 (Q) \rightarrow (1)$

(R) Electrostatics binding energy $\propto Z(Z-1)$

$(R) \rightarrow (2)$

(S) For stable nuclei with mass no. in range 30 to 170 binding energy per nucleon is constant and graph is straight line, So, $(S) \rightarrow (4)$

43. Inside the sphere field varies linearly i.e., $E \alpha r$ with distance and out side varies according to $E \alpha \frac{1}{r^2}$ Hence the variation is shown by curve (1)

44. Initially $X_e = R_1 X_L = 2R$

$X_e^1 = 4R X_L^1 = 2X_L, R^1 = 2R$

$X_e^1 - X_L^1 = 0$

$\varnothing = 0$

45. $\ell = 10\text{cm} = 0.1\text{m}, d = 0.2\text{mm},$

$e = 0.2\ T = 2000K$

$E = \sigma(2\pi r\ell)eT^4 = 11.4W$

46. resistance of bulb, $R_{bulb} = \frac{V^2_{bulb}}{p_{bulb}} = 0.5\Omega$

In order to the bulb glows at full intensity, current will pass through the bulb $= \frac{4.5W}{1.5V} - 3A$

Equivalent resistance of circuit, $R_{eq} = \frac{5}{3} + \frac{0.5\times1}{0.5+1} = \frac{5}{3} + \frac{0.5}{1.5} = \frac{5}{3} + \frac{1}{3} = 2\Omega$

$\therefore$ current supplied by battery $\frac{E}{R_{eq}} = \frac{E}{2}$

Current in bulb $= I \times \frac{1}{(1+0.5)} \Rightarrow 3A = \frac{I}{1.5} = \frac{E}{2\times1.5} \Rightarrow E = 9\text{volt.}$

47. Takingn the velocity w.r.t train, from equation of continuity we get

$3S_1 v_t = (3S_t - S_t)v$

$\Rightarrow v = \frac{3}{2}v_t$

According to Bernoulli's equation, we get

$\Rightarrow p_0 + \frac{1}{2}pv_t^2 = p + \frac{1}{2}\rho\left(\frac{9}{4}v_t^2\right)$

$\Rightarrow p_0 - p = \frac{5\rho v_t^2}{8} = \frac{5}{2(4)}\rho v_t^2$

48. Mean free path $\lambda = \frac{1}{\sqrt{2}\pi d^2 n}$

n= number of molecules per unit volume, d= diameter of molecule as the condition for both gases are identical so n will be constant $\lambda \alpha \frac{1}{d^2} \lambda_1 : \lambda_2 = 4:1$

49. B catches C in time t then $t = \frac{d}{u-10}$

Seperation by this time has increased by 'd' between A and C hence

$(10-5)\times\frac{d}{(u-10)} = d$

$u = 15m/s$

50. $9 = v_1 + 2v_2$

$v_2 - v_1 = 9(\text{elastic collision})$

$v_2 = 6m/s$

$2m \times 6 = 3mv'$

$v' = 4m/s$

CHEMISTRY

51. For Lyman series

$$\Delta\bar{v} = \bar{v}_{max} - \bar{v}_{min} = R\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right) - R\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = R - \frac{3R}{4} = \frac{R}{4}$$

For Balmer series,

$$\Delta\bar{v} = \bar{v}_{max} - \bar{v}_{min} = R\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right) - R\left(\frac{1}{2^2} - \frac{1}{3^2}\right)$$

$$= \frac{R}{4} - \frac{5R}{36} = \frac{4R}{36} \qquad \therefore \frac{\Delta\bar{v}_{Lyman}}{\Delta\bar{v}_{Balmer}} = \frac{\frac{R}{4}}{\frac{4R}{36}} = \frac{36}{16} = 9:4$$

52. Dipole moment is a vector quantity and by convention it is depicted by a small arrow with tail on positive centre and head pointing towards the negative center. Hence statement-I is incorrect. The crossed arrow symbolizes the direction of the shift of electron density (charges) in the molecule hence, statement-II is correct.

53. $A_{(g)} \rightleftharpoons B_{(g)} + \frac{C_{(g)}}{2}$

At $t = 0$ a 0 0

At $t = t_{eq}$ $a - x$ x $x/2$

For a mole, x moles are dissociated.

For 1 mole, $\frac{x}{a}$ moles $= \alpha$ moles are dissociated

$\therefore x = a\alpha$ $\qquad A_{(g)} \rightleftharpoons B_{(g)} + \frac{C_{(g)}}{2}$

At $t = t_{eq}$ $a - a\alpha$ $a\alpha$ $\frac{a\alpha}{2}$

Total number of moles at equilibrium $a + \frac{a\alpha}{2} = a\left(1 + \frac{\alpha}{2}\right)$

$$P_A = \frac{a(1-\alpha)P}{a\left(1+\frac{\alpha}{2}\right)}, P_B = \frac{a\alpha.P}{a\left(1+\frac{\alpha}{2}\right)}, Pc = \frac{\frac{a\alpha}{2}.P}{a\left(1+\frac{\alpha}{2}\right)}$$

$$P_A = \frac{(1-\alpha)P}{1+\frac{\alpha}{2}}, P_B = \frac{\alpha P}{1+\frac{\alpha}{2}}, Pc = \frac{\left(\frac{\alpha}{2}\right)P}{1+\frac{\alpha}{2}} \qquad K_P = \frac{P_B.(P_C)^{1/2}}{P_A} = \frac{\left(\frac{\alpha P}{1+\frac{\alpha}{2}}\right)\left(\frac{\frac{\alpha P}{2}}{1+\frac{\alpha}{2}}\right)^{1/2}}{\frac{(1-\alpha)P}{\left(1+\frac{\alpha}{2}\right)}}$$

$$K_P = \frac{\alpha.\alpha^{1/2}.P^{1/2}}{(2+\alpha)^{1/2}(1-\alpha)} = \frac{\alpha^{3/2}.P^{1/2}}{(2+\alpha)^{1/2}(1-\alpha)}$$

54. Given reaction,

$$Fe^{2+}_{(aq)} + Ag^{+}_{(aq)} \rightarrow Fe^{3+}_{(aq)} + Ag_{(s)}$$

$$Fe^{2+}_{(aq)} + 2e^- \rightarrow Fe_{(s)}; E^0_{Fe^{2+}/Fe} = yV.....(i)$$

$\Delta G_1^0 = -2Fy$

$Fe^{3+} 3e^- \rightarrow Fe_{(s)}; E^0_{Fe^{3+}/Fe} = zV.....(ii)$

$\Delta G_2^0 = -3Fz$

$Ag^+ + e^- \rightarrow Ag_{(s)}; E^0 = xV.....(iii)$

$\Delta G_3^0 = -Fx$

On adding equations (i) and (iii) and subtracting (ii) from it, we get

$Fe^{2+}_{(aq)} + Ag^+_{(aq)} \rightarrow Ag_{(s)} + Fe^{3+}_{(aq)}$

$\therefore \Delta G = \Delta G_1^0 + \Delta G_3^0 - \Delta G_2^0 = -FE^0{}_{cell}$

$= -2Fy - Fx - (-3Fz)$

$-E^0{}_{cell} = -2y - x + 3z$ or $E^0_{cell} = x + 2y - 3z$

55. $[Xe]4f^4 6s^2$

56. As the atomic size increases down the group, the bond length increases and hence the bond strength decreases. Consequently, the cleavage of E-H bond (where E is element of group 16) Becomes easier. As a result the tendency to release hydrogen as proton increases i.e., acidic strength increases down the group. Thus, the order of acidic strength is $H_2O < H_2S < H_2Se < H_2Te$

57. $2MnO_2 + 4KOH + O_2 \rightarrow \underset{\text{Green (B)}}{2K_2MnO_4} + 2H_2O$

$3K_2MnO_4 + 4HCl \rightarrow \underset{\text{Purple (C)}}{2KMnO_4} + MnO_2 + 2H_2O + 4Kcl$

$\underset{(C)}{2KMnO_4} + H_2O + KI \rightarrow 2KOH + \underset{(A)}{2MnO_2} + \underset{(D)}{KIO_3}$

58. In $[Ti(H_2O)_6]^{2+}$; Ti is in +2 oxidation state. Hence, its electronic configuration is $t_{2g}^2 e_g^0$ as H_2O does not cause pairing of electrons. CFSE $= [-0.4(2) + 0.6(0)]\Delta_0 = -0.8\Delta_0$

In $[V(H_2O)_6]^{2+}$; V is in +2 oxidation state. Hence its electronic configuration is $t_{2g}^3 e_g^0$ as H_2O does not cause pairing of electrons

CFSE=$[-0.4(3) + 0.6(0)]\Delta_0 = -1.2\Delta_0$

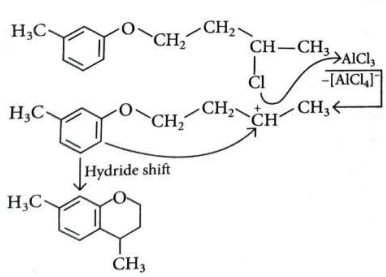
In $[Mn(H_2O)_6]^{3+}$, Mn is in +3 oxidation state. Hence electronic configuration is $t_{2g}^3 e_g$ as H_2O does not cause pairing of electrons. CFSE $= [-0.4(3) + 0.6(1)]\Delta_0 = -0.6\Delta_0$

In $[Fe(H_2O)_6]^{3+}$: Fe is in +3 oxidation state. Hence, its electronic configuration is $t_{2g}^3 e_g^2$

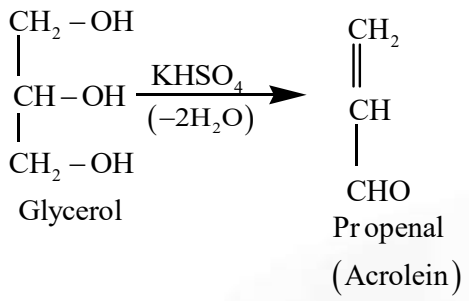
CFSE=$[-0.4(3) + 0.6(2)]\Delta_0 = (-1.2 + 1.2)\Delta_0 = 0$

Thus, A – IV, B – I, C – II, D – III

59.



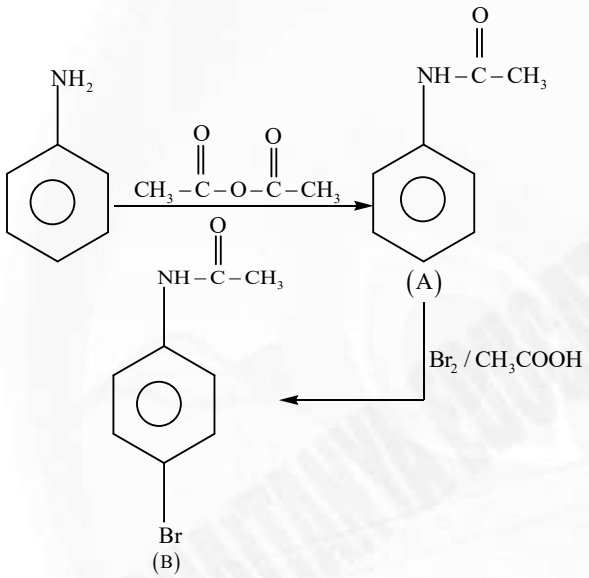
60.



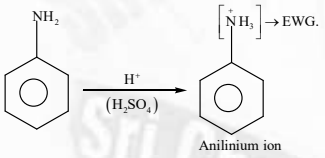
Acrolein has pungent smell instead of fruity odour.

61. The given reaction is known as Gattermann-Koch reaction.

62.



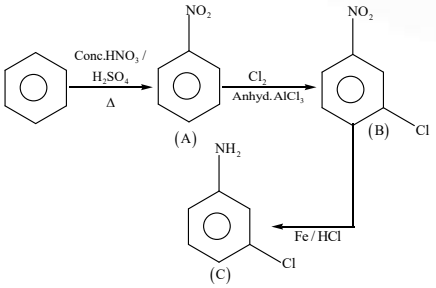
63. In aniline, amine group is protonated in strongly acidic medium to from anilinium ion which acts as meta directing group. Besides ortho and para product, some amount of meta product is also formed.



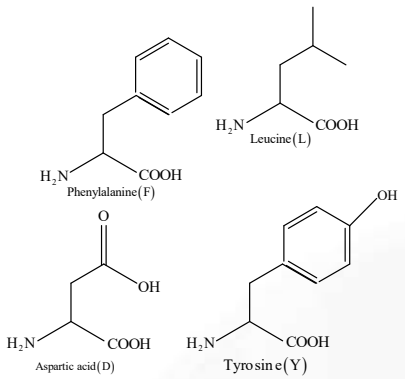
The order of percentage yield is as follow: C>B>A

64. Gabriel phthalimide synthesis is used for preparation of aliphatic primary amines. Aromatic primary amines cannot be prepared as aryl halides don not undergo nucleophilic substitution.

65.



66.



Arranging these four peptides we will get the given tetra peptide. thus the tetra peptide is FLDY.

67. Molecular mass of $H_{12}O_6Cl_3Cr = 265$

On treatment with conc. H_2SO_4, it loses 13.5% of its original mass.

$\therefore$ Loss of mass $= \frac{265 \times 13.5}{100} = 35.78; 36 \equiv 2$ mole of H_2O

Thus, two H_2O molecules are lost. Hence, molecular formula of the complex is $\left[Cr(H_2O)_4 Cl_2\right]Cl.2H_2O$.

68. Radial node/spherical node=*n-l*-1

For 2s-orbital,n=2,l=0, then radial nodes=2-0-1=1

Only graph (2) has one node.

69. $\lambda_m = \lambda_m^0 - b\sqrt{c}$

70. Molality (m)

$$= \frac{x_2 \times 1000}{x_1 \times M_1} = \frac{0.1 \times 1000}{0.9 \times M_1} = \frac{1000}{9M_1} \qquad \Rightarrow M_1 = \frac{1000}{9m}$$

$$\text{Also, } m = \frac{1000M}{1000d - M_2 \times M} \qquad \Rightarrow \frac{m}{M} = \frac{1000}{1000 \times 2 - M_2 \times M}$$

$$\Rightarrow 1 = \frac{1000}{2000 - M_2 \times M} \quad [\because m = M \text{ given}]$$

$$\Rightarrow M \times M_2 = 1000 \Rightarrow M_2 = \frac{1000}{M}$$

On dividing equation (ii) by (i), we get

$$\frac{M_2}{M_1} = \frac{1000}{M} \times \frac{9m}{1000} \qquad \Rightarrow \frac{M_2}{M_1} = 9$$

$$[\because m = M] \qquad \text{or } \frac{M_{solute}}{M_{solvent}} = 9$$

71. $Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \rightarrow 6Fe^{3+} + 2Cr^{3+} + 7H_2O$

1 mole of $Cr_2O_7^{2-}$ requires 6 moles of Fe^{2+}

$$6 \times \left(\frac{20 \times 0.03}{1000}\right) = \frac{x \times 15}{1000} \Rightarrow x = \frac{6 \times 20 \times 0.03}{15} = 0.24$$

$x = 24 \times 10^{-2}$ M

72. $\Delta G = \Delta H - T\Delta S$

At equilibrium, $\Delta G = \Delta H - T\Delta S$

$$T = \frac{\Delta H}{\Delta S} = \frac{-165\times10^{3}\,J/mol}{-550JK^{-1}} = 300K$$

73. $k = Ae^{-\frac{E_a}{RT}}$

$$K_{T,catalysed} = Ae^{-\frac{E_a'}{RT}}; K_{T\,uncatalysed} = Ae^{-\frac{E_a}{RT}}$$

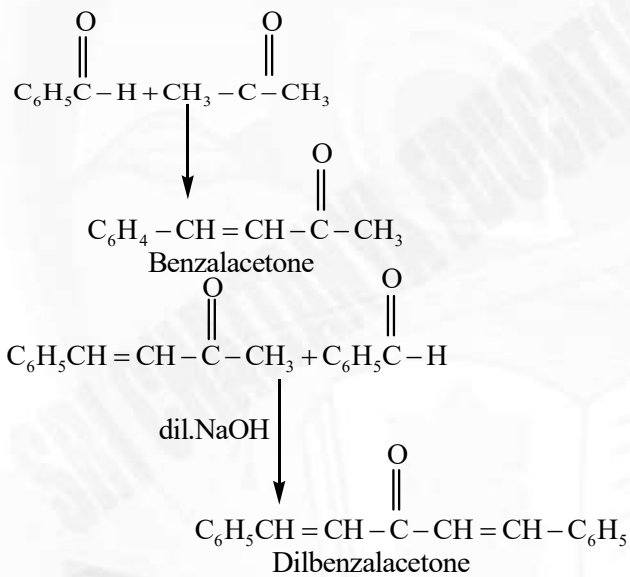
$$\frac{k_{T,catalysed}}{k_{T\,uncatalysed}} = e^{\frac{E_a - E_a'}{RT}} = e^{\frac{10\times1000}{8.31\times300}} = e^{4} \therefore x = 4$$

74. $R_f\,value = \frac{Distance\ travelled\ by\ solute}{Distance\ travelled\ by\ solvent}$

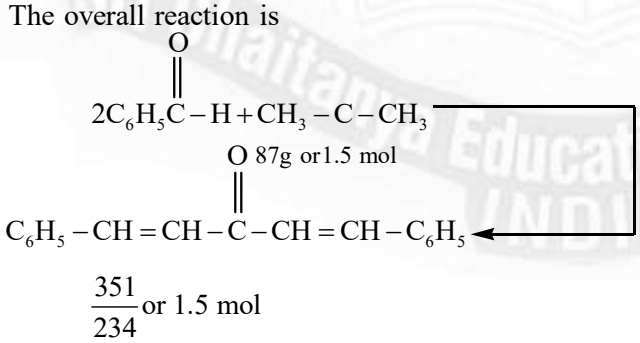
$$R_f(A) = \frac{5-1}{8} = \frac{4}{8}\quad R_f(B) = \frac{7-1}{8} = \frac{6}{8}$$

$$\frac{R_f(B)}{R_f(A)} = \frac{6}{8}\times\frac{8}{4} = 1.5 = 15\times10^{-1}$$

75. Base catalyzed crossed aldol condensation between and aromatic aldehyde and aliphatic aldehyde or a ketone is Claisen-Schmidt condensation.



The overall reaction is



For 1.5 mole acetone, benzaldehyde required

$= 1.5\times2 = 3$ mol

$= 3\times106g = 318g$